

Comparative—Quantitative Investigations on Brains of Feral Pigs from the Galapagos Islands and of European Domestic Pigs*

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Summary. An intraspecific allometric comparison between the brain weights of feral pigs from the Galapagos Islands and those of European domestic pigs shows identical brain sizes in relation to the body sizes. The return of domestic pigs to the wild life state some generations ago has produced no increase in brain size. Hence this process is not reversible to the process of domestication, domestic pigs having 33% smaller brains than free living mutants of their ancestral species the European wild boar (*Sus scrofa scrofa* L.).

However, quantitative investigation shows a partly different composition of the brains of feral pigs compared with those of European domestic pigs. Of the fundamental brain parts, telencephalon and mesencephalon of the feral pigs are not different in volume size, whereas diencephalon and medulla oblongata are 6.5% and 10% larger, respectively; the cerebellum on the contrary is about 11% smaller than in domestic pigs. A further subdivision of the telencephalon into the regions of neocortex, corpus striatum and allocortex, and that of the allocortex into some olfactory and limbic centers, elucidates other differences. A discussion of these results under functional aspects and in connection with the process of return to wild-life conditions is given.

Key words: Pig — Brain — Feral state.

Introduction

The successful feral state of domestic animals throughout several generations is a special zoological phenomenon. Man formed the domestic animals which, in many characteristics, have changed due to the new and special conditions of domestication. This can be proved by comparing domestic animals with the corresponding species of their mutual ancestors still living under natural conditions (Herre and Röhrs, 1971a, 1973). Through accidental return to wild-life state, domestic animals may secondarily be submitted to the natural environment. In this case it is of interest to learn more fully about the possibility, degree and causes of adaption. The present state of knowledge concerning the changes of domestic animals after their return to wild-life state has repeatedly been summarized by Herre and Röhrs (1971a, b, 1973). Considerable uncertainty owing to dearth of investigations is evident. This study should therefore serve to amplify the present scanty state of information.

* The term feral is a synonym for the German word "verwildert". It means non-indigenous, free-living or wild animals originating from domestic stock. This definition was already used by Pullar (1950), Letts (1964), Shank (1972) and others.

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The central nervous system is in this connection an organ of special interest, since the ability of adaption to new environments depends to a great degree on its efficiency. The size of the brain or its parts can—critically used—serve as a graduator for its efficiency, i.e. be considered a criterion of its functional ability. If a brain structure is enlarged or reduced within a certain derivation line an adequate functional increase or decrease, respectively, of this region can be assumed. Thus accomplished comparative-quantitative investigations of the brains of European wild boars (*Sus scrofa scrofa* L.) and of those of European domestic pigs (*Sus scrofa* f. dom.) have shown that the brains have decreased in size to about $\frac{1}{3}$ due to domestic conditions. But in this very strong reduction of total brain size, different and considerable intensities of reductions are included with regard to various brain parts. Thus the neocortex with 38.5%, the entire allocortical limbic structures with 40.7% and some visual brain centers with 41.0% were seen to be subjected to especially strong reductions. Compared with this, the complete olfactory brain structures with 32.2% seem very remarkable because of their small decrease (Kruska, 1970, 1972; Kruska and Stephan, 1973; Röhrs and Kruska, 1969).

Pigs on the Galapagos Islands returned to the wild-life state at least 70, on some islands even 140 years ago. In this study the quantitative composition of the brains of such feral pigs is ascertained and compared with that of the domestic pigs. The possible importance of the results will be discussed.

Material and Methods

On the Galapagos journey Herre/Röhrs in September 1971, eight feral pigs were shot on the island of Santiago. These animals were prepared. The body weights, namely gross weights as well as net weights (= gross weight minus weight of viscera) and the fresh brain weights were ascertained. These data are summarized in Table 1. The brains were fixed and preserved in a 4% formalin solution.

The four best preserved brains were cut into serial sections at equal intervals, and the fresh volume sizes of several brain parts were ascertained (Table 2) using enlarged photographs. These methods have been described in previous publications (Kruska, 1970; Kruska and Stephan, 1973; Stephan, 1960, 1966; Stephan *et al.*, 1970). Former published values of six domestic pig brains, likewise obtained (Kruska, 1970; Kruska and Stephan, 1973), are re-summarized in Table 3.

Comparisons of brain weights must be accomplished by means of the allometrical methods because only thus is the influence of the body size on brain size eliminated (Röhrs, 1961). As declared below, and as already noted (Herre and Röhrs, 1973; Kruska, 1973a), the total brain size of feral pigs shows no increase in the substance. Nevertheless we have carried out a comparative analysis to obtain information on the proportional relationship within the brains.

When comparing two or more groups of animals belonging to the same species, changes or differences in the sizes of brain parts are calculable by means of the relative values when it is evident to what degree the total brain size differs. In a preliminary study we have made a detailed report on this fact (Kruska and Stephan, 1973). In this study, the average proportional differences of the feral pig brains are to be ascertained on the basis of the brain structure of European domestic pigs. These differences in the size of the various brain structures can be calculated according to the following formula:

$$SD_{\text{brain part}} = \frac{RV_{\text{feral}} \times (SD_{\text{total brain}} + 100)}{RV_{\text{domestic}}} - 100.$$

SD = size difference percentage

RV = relative value

($\times$ = multiplication sign)

As no difference in total brain size between feral pigs and domestic pigs was ascertained in this special case (see below), the size differences of the various brain structures must be comprehended according to the formula:

$$SD_{\text{brain part}} = \frac{RV_{\text{feral}} \times 100}{RV_{\text{domestic}}} - 100.$$

The individual percentages of the measured brain structures in the total brain sizes of feral pigs are summarized in Table 4 and those of the domestic pigs in Table 5. Comparative reflections yield information on the individual variability of the proportional brain structure.

When estimating the size differences of the various structures, we started from the absolute volume sizes (Tables 2 and 3) and ascertained the geometrical mean for each structure (Table 6, Columns 1 and 4). The geometrical variability coefficients (Table 6, Columns 2 and 5) are gradators for the variation degree of the structure sizes within both groups. The relative values, needed for calculation, are the geometrical means percentages of the various structures on the geometrical mean of the total brains (Table 6, Columns 3 and 6). The ascertained size differences of the measured structures between feral and domestic pig brains can be seen in Table 6, Column 7. With the signs plus and minus is indicated whether a structure of the feral pigs is larger or smaller than that of the domestic pigs. These calculated differences serve as value indicators. Of course, they allow no precise orientation but are nevertheless able to indicate the important size differences. Further addition of material may modify these indicators, but remarkable differences become evident.

Furthermore, we have accomplished a calculation of the size differences of the brain structures between feral and domestic pigs through the arithmetical means of the individual relative values—as indicated in Tables 4 and 5—and their standard deviation, as a measurement for the individual variability in size. As a result we obtained values almost identical to those attained in the calculation methods submitted here. Therefore we need no further statements.

Results

First of all we must compare the total brain sizes of the feral Galapagos pigs with those of the European domestic pigs. This is done by means of the allometrical methods. The intraspecific relationship between brain and body weight of the domestic pigs is the basis for comparison. Already in a former study (Kruska and Stephan, 1973), we were able to calculate the intraspecific allometric line for the weight data of 116 domestic pigs:

$$\log \cdot y = 0.6371 + 0.30 \times \log \cdot x$$

$y = \text{brain weight}; x = \text{body weight}$
($\times$ = multiplication sign)

With a value of 0.30 the allometric exponent 'a' indicates the intraspecific relationship between brain size and body size as being typical for mammals. Because of the substantial fat portion in the bodies of domestic pigs, we took the net body weights as a relation-basis. The gross body weights had to be reduced by 45%, namely by 10% for the viscera and by 35% for the fat.

The feral Galapagos pigs have not put on any fat (Herre and Röhrs, 1973). Accordingly, the net body weights from Table 1 (i.e. the gross weights minus the viscera weights) must be regarded as comparable values in relation to the domestic net weights. Previously, the body weights of European wild boars were similarly dealt with (Kruska, 1970; Kruska and Stephan, 1973).

In this connection, the differences between the two body weight data for feral pigs in Table 1 are notable. Our calculations have demonstrated that the net weight percentage of the gross weight varies from 70.7% (A 21) to 78.9% (A 26)

Table 1. Data on the feral pigs of the Galapagos Islands

No.	Date	Sex	Age	Body weight in g		Brain weight in g
				gross	net	
A 21	3. 9. 71	♂	ad.	43 400	30 700	99.6
A 22	3. 9. 71	♂	ad.	55 450	42 400	110.5
A 23	4. 9. 71	♂	ad.	43 300	33 600	98.5
A 26	7. 9. 71	♂	ad.	62 100	49 000	106.2
A 27	6. 9. 71	♂	ad.	56 450	43 900	102.5
A 28	6. 9. 71	♂	subad.	34 600	26 000	90.0
A 29	6. 9. 71	♂	ad.	51 500	40 400	100.0
A 30	7. 9. 71	♂	subad.	31 900	24 000	87.2

and that the average of all animals amounts to 76.3%. I.e. the viscera weight of a feral Galapagos pig is $\frac{1}{4}$ of the total body weight. The individual variability is thereby limited to $\pm 5\%$.

For the relationship between brain weight and body weight of the feral pigs calculations were made, although only values for eight animals exist. For the three allometric lines as regression line, reduced main axis of the distribution ellipsis and main axis of the distribution ellipsis, 'a'-values of 0.2751, 0.3041 and 0.2793 have been determined. The correlation of the individual values to these lines is statistically assured with 0.1%. Consequently, the typical intraspecific allometry is here, also, rather outstanding. The relationship between brain and body weight has not changed because of the influence of feral life. We can verify the 'a'-value of 0.30 as being typical for pigs and boars; and for the feral pigs we obtain the intraspecific allometric line:

$$\log \cdot y = 0.6317 + 0.30 \times \log \cdot x$$

($\times$ = multiplication sign)

Fig. 1 shows the allometric lines of both domestic and feral pigs in a double logarithmic scale, and the values of the eight feral pigs.

A comparison of the integration constancy ('b'-value) of the feral pigs with that of the domestic pigs mentioned above yields a non-significant low difference (on the average, the eight feral pigs have 1.2% less total brain weight than the domestic pigs of comparable body weight). This difference is doubtlessly due to a too small amount of material and to the individual variability. We accept as a fact that the feral pigs have the same brain size as the European domestic pigs. This finding is in accordance with the results obtained on other domesticated animals which successfully returned to wild-life state throughout longer periods of time (Herre and Röhrs, 1973).

Now the question arises whether the sizes of the various brain parts, i.e. the proportional construction of the brains of domestic and feral pigs, are likewise equal. From Table 6, Column 7, it can be concluded that the examined brain regions of feral pigs are to a greater degree larger or smaller than those of the domestic pigs. A certain difficulty in the interpretation of these findings is caused by the limit of significancy, since in view of the small amount of material no level

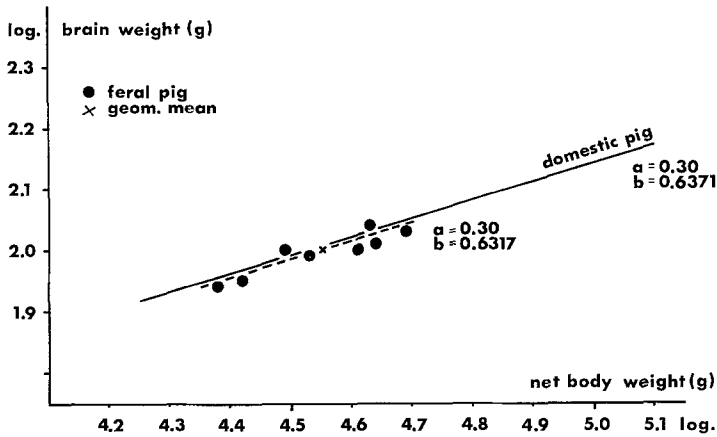


Fig. 1. The relationship between brain weight and body weight of feral Galapagos pigs in a double logarithmic scale. Reference base is the intraspecific allometry line of the domestic pigs (see text)

Table 2. Data on volume sizes of the fundamental brain sections, the allocortical subdivisions, the total brains (in mm³) and brain weights (in g) of the feral pigs from the Galapagos Islands

	A 23	A 21	A 26	A 22
Medulla oblongata	6395	6678	6717	7240
Cerebellum	11532	10419	11408	11153
Mesencephalon	4010	4193	4473	4615
Diencephalon	6939	6528	7104	7607
Telencephalon	62946	63989	69174	71931
Neocortex	47192	48477	52320	53229
Corpus striatum	3335	3423	3671	3779
Allocortex	12419	12089	13184	14923
Bulbus olfactorius	887 ^a	1147 ^a	1542	1650
Regio retrobulbaris	406	356	432	447
Regio praepiriformis	2933	2766	3054	3305
Amygdaloid complex	1061	1113	1192	1248
a) centromedial group	414	410	440	478
b) basolateral group	647	703	752	769
Tuberculum olfactorium	721	686	707	836
Nuclei basalis	753	829	773	939
Palaeocortex + Amygdaloid complex	5874	5750	6158	6775
Septum	674	695	728	787
Hippocampus	3852	3456	3675	4508
Schizocortex	1131	1042	1081	1203
limbic structures	5657	5193	5484	6498
brain weight	98.5	99.6	106.2	110.5
brain volume	95077	96139	102510	106660

^a Damaged olfactory bulb.

Table 3. Data on volume sizes of the fundamental brain sections, the allocortical subdivisions, the total brains (in mm³) and brain weights (in g) of the domestic pigs

	Sd 5	Sd 9	Sd 31	Sd 19	Sd 17	Sd 12
Medulla oblongata	5207	6431	6656	6755	6927	7158
Cerebellum	11477	13953	12299	14614	13322	14203
Mesencephalon	3627	4137	4390	4458	4733	5175
Diencephalon	6319	6628	7151	7153	7187	7694
Telencephalon	58703	66335	69415	71071	77356	82386
Neocortex	44826	50018	53362	53488	60449	64205
Corpus striatum	3386	3975	3706	3815	3930	4573
Allocortex	10490	12342	12346	13767	12977	13608
Bulbus olfactorius	835 ^a	1263	929 ^a	1302	1408	1214 ^a
Regio retrobulbaris	342	423	418	444	430	405
Regio praepiriformis	2420	2799	3085	3451	2935	3252
Amygdaloid complex	1056	1101	1216	1217	1154	1322
a) centromedial group	395	451	457	463	441	552
b) basolateral group	661	650	759	754	713	770
Tuberculum olfactorium	775	863	903	1103	961	1280
Nuclei basalis	589	655	626	836	763	750
Palaeocortex + Amygdaloid complex	5182	5840	6248	7051	6242	7009
Septum	628	753	772	864	808	952
Hippocampus	2943	3366	3314	3372	3353	3257
Schizocortex	900	1118	1080	1177	1164	1173
limbic structures	4473	5237	5168	5413	5326	5383
brain weight	92.0	105.0	110.0	112.0	119.0	125.0
brain volume	88803	101351	106178	108108	114865	120656

^a Damaged olfactory bulb.

of significance can be achieved. Accordingly, we decided to take into consideration the individual variations of the brain structure sizes (although rather unimportant) and only to interpret those size differences exceeding 5% as real differences. Structures reaching only small values of difference between feral and domestic pigs have been regarded as being equal in size. According to such aspects, telencephalon and mesencephalon of the feral pigs are not different in volume size from those of domestic pigs; diencephalon and medulla oblongata are with 6.5% and 10% respectively, insignificantly larger; the cerebellum on the contrary is about 11% smaller.

A further classification of the telencephalon demonstrates, however, that in spite of the total size being the same, the allocortex of the feral pigs is 11% larger, whereas neocortex and corpus striatum show almost the same volume. But within the allocortex the bulbus olfactorius becomes especially significant as in the undamaged samples it is 28% larger (including the damaged specimens the difference is still 18%). Also the hippocampus formation and the nuclei basalis are noticeable through the evidently clear size differences. In feral pigs these structures are about $\frac{1}{4}$ larger than in domestic pigs.

Table 4. Brain divisions and subdivisions percentages of the brains of feral pigs from the Galapagos Islands. Total Brain volume = 100%

	A 23	A 21	A 26	A 22
Medulla oblongata	6.7261	6.9462	6.5525	6.7879
Cerebellum	12.1291	10.8374	11.1287	10.4566
Mesencephalon	4.2176	4.3614	4.3635	4.3268
Diencephalon	7.2983	6.7902	6.9301	7.1320
Telencephalon	66.2053	66.5588	67.4802	67.4395
Neocortex	49.6356	50.4239	51.0389	49.9053
Corpus striatum	3.5077	3.5605	3.5811	3.5430
Allocortex	13.0620	12.5745	12.8612	13.9912
Bulbus olfactorius	0.9329 ^a	1.1931 ^a	1.5042	1.5470
Regio retrobulbaris	0.4270	0.3703	0.4214	0.4191
Regio praepiriformis	3.0849	2.8771	2.9792	3.0986
Amygdaloid complex	1.1159	1.1577	1.1628	1.1701
a) centromedial group	0.4354	0.4265	0.4292	0.4482
b) basolateral group	0.6805	0.7312	0.7336	0.7210
Tuberculum olfactorium	0.7583	0.7136	0.6897	0.7838
Nuclei basalis	0.7920	0.8623	0.7541	0.8804
Palaeocortex + Amygdaloid complex	6.1782	5.9809	6.0072	0.3520
Septum	0.7089	0.7229	0.7102	0.7379
Hippocampus	4.0515	3.5948	3.5850	4.2265
Schizocortex	1.1896	1.0838	1.0545	1.1279
limbic structures	5.9499	5.4016	5.3497	6.0923

^a Damaged olfactory bulb.

In other respects the palaeocortical structures must be considered as being of the same size or only unimportantly larger, in total (including the amygdaloid complex) as well as in separate. Only the tuberculum olfactorium is 19% smaller, while the region of the basal nuclei is 24.8% larger.

In relation to the olfactory structures of the allocortex, the total limbic structures of this brain part are 17% larger. This great difference is due mainly to the size of the hippocampus formation, which, as already mentioned, is 25.4% larger. Likewise the schizocortex is unimportantly larger, whereas the septum of feral and domestic pigs must be regarded as being of the same size.

Furthermore, the results of the individual variability are worth mentioning. Already the comparison of the relative values of the measured structures of the feral pigs (Table 4) with those of the domestic pigs (Table 5) demonstrates that the individual variation breadth in the relative sizes of the various regions in feral pigs is not that large. Most of the percentage-values of the animals lie closer together than those of the domestic pigs.

However, the finding just described becomes even more significant through the geometrical variability coefficients of the brain parts (Table 6, Columns 2 and 5). For the vast majority of the structures, this graduator is significantly smaller in the feral pigs than in the domestic pigs. Exceptions to this rule, however, exist. Thus, for instance, the regio retrobulbaris (= nucleus anterior) and

Table 5. Brain divisions and subdivisions percentage of the brains of domestic pigs. Total brain volume = 100%

	Sd 5	Sd 9	Sd 31	Sd 19	Sd 17	Sd 12
Medulla oblongata	5.8635	6.3453	6.2687	6.2484	6.0306	5.9326
Cerebellum	12.9241	13.7670	11.5834	13.5180	11.5980	11.7715
Mesencephalon	4.0843	4.0819	4.1346	4.1237	4.1205	4.2891
Diencephalon	7.1158	6.5396	6.7349	6.6165	6.2569	6.3768
Telencephalon	66.1047	65.4508	65.3761	65.7407	67.3451	68.2817
Neocortex	50.4780	49.3513	50.2571	49.4764	52.3261	53.2133
Corpus striatum	3.8129	3.9220	3.4904	3.5289	3.4214	3.7901
Allocortex	11.8217	12.1775	11.6276	12.7345	11.2976	11.2783
Bulbus olfactorius	0.9403 ^a	1.2462	0.8749 ^a	1.2044	1.2258	1.0062 ^a
Regio retrobulbaris	0.3851	0.4174	0.3937	0.4107	0.3744	0.3357
Regio praepiriformis	2.7251	2.7617	2.9055	3.1922	2.5552	2.6953
Amygdaloid complex	1.1891	1.0863	1.1452	1.1257	1.0047	1.0957
a) centromedial group	0.4448	0.4450	0.4304	0.4283	0.3839	0.4575
b) basolateral group	0.7443	0.6413	0.7148	0.6975	0.6207	0.6382
Tuberculum olfactorium	0.8727	0.8515	0.8505	1.0203	0.8366	1.0609
Nuclei basalis	0.6633	0.6463	0.5896	0.7733	0.6643	0.6216
Palaeocortex + Amygdaloid complex	5.8354	5.7622	5.8845	6.5222	5.4342	5.8091
Septum	0.7072	0.7430	0.7271	0.7992	0.7034	0.7890
Hippocampus	3.3141	3.3211	3.1212	3.1191	2.9191	2.6994
Schizocortex	1.0135	1.1031	1.0172	1.0887	1.0134	0.9722
limbic structures	5.0370	5.1672	4.8673	5.0070	4.6367	4.4614

^a Damaged olfactory bulb.

the basolateral group of the amygdaloid complex of the feral pigs have an unimportantly higher geometrical variability coefficient than the same regions of the domestic pigs. However, the very high coefficient of the hippocampus formation, amounting to 12.0 in the feral pigs, is especially remarkable compared with the very low coefficient of 5.4 in the domestic pigs. This unusually high individual variation breadth of the hippocampus size in the feral pigs also becomes noticeable when summarizing all measured limbic structures, so that here also the geometrical variability coefficient is larger than that of the domestic pigs. The higher coefficient for bulbus olfactorius of the feral pigs must be attributed to the inclusion of damaged samples. When considering only undamaged bulbs, the variability coefficient of the feral pigs is smaller than that of the domestic pigs.

This finding on the individual variability of the structure sizes in the examined material may at first seem unimportant. However, it becomes interesting when the similarly obtained results from brains of European wild boars are taken into consideration (Kruska, 1972; Kruska and Stephan, 1973). Also, in comparison with brains of domestic pigs, it became clear that fundamentally the individual size variability of the structures in wild boars is more limited than the variability of the domestic pigs. Ordinarily the geometrical variability coefficient of the wild animals is significantly smaller. But one can find exceptions here too. For example,

Table 6. Data on geometrical means, the geometrical variability coefficients, the divisions and subdivisions percentages (geometrical means) of total brains of domestic pigs and feral pigs, and the percentage differences of the various structures between domestic and feral pigs

	Domestic pigs ($n = 6$)			Feral pigs ($n = 4$)			(7)
	(1)	(2)	(3)	(4)	(5)	(6)	
Medulla oblongata	6489.0	12.0	6.1122	6750.7	5.3	6.7516	+10.46%
Cerebellum	13264.0	9.8	12.4938	11120.0	4.7	11.1216	-10.98%
Mesencephalon	4393.6	12.8	4.1385	4316.3	6.5	4.3169	+ 4.31%
Diencephalon	7008.1	7.2	6.6012	7034.0	6.5	7.0350	+ 6.57%
Telencephalon	70480.0	12.6	66.3874	66908.0	6.5	66.9174	+ 0.80%
Neocortex	54016.0	13.8	50.8794	50241.0	6.0	50.2480	- 1.24%
Corpus striatum	3881.5	10.4	3.6561	3547.5	6.0	3.5480	- 2.96%
Allocortex	12539.0	10.4	11.8109	13110.0	9.8	13.1118	+11.00%
Bulbus olfactorius	1138.7	23.1	1.0726	1268.4	33.1	1.2686	+18.27%
Bulbus olfactorius ^a	1322.9	5.8	1.2461	1595.0	4.9	1.5952	+28.02%
Regio retrobulbaris	408.9	9.7	0.3852	408.7	10.6	0.4088	+ 6.13%
Regio praepiriformis	2971.4	13.3	2.7989	3008.2	7.8	3.0086	+ 7.49%
Amygdaloid complex	1174.5	8.4	1.1063	1151.3	7.5	1.1515	+ 4.09%
a) centromedial group	457.6	11.4	0.4310	434.7	7.3	0.4348	+ 0.88%
b) basolateral group	716.2	7.6	0.6746	716.2	8.1	0.7163	+ 6.18%
Tuberculum olfactorium	967.4	19.8	0.9112	735.3	9.2	0.7354	-19.29%
Nuclei basalis	697.9	14.4	0.6574	820.4	10.4	0.8205	+24.81%
Palaeocortex + Amygdaloid complex	6227.5	12.3	5.8659	6126.9	7.6	6.1278	+ 4.46%
Septum	789.7	15.1	0.7438	719.8	7.0	0.7199	- 3.21%
Hippocampus	3263.9	5.4	3.0744	3853.7	12.0	3.8542	+25.36%
Schizocortex	1097.4	10.8	1.0337	1112.7	6.4	1.1129	+ 7.66%
limbic structures	5156.0	7.4	4.8566	5688.1	10.0	5.6889	+17.14%
brain weight	110.0	11.3		103.6	5.6		
brain volume	106164.8	11.3	100.0000	99986.0	5.6	100.0000	0%

^a Only undamaged olfactory bulbs (domestic pigs $n = 3$, feral pigs $n = 2$).

Columns (1) and (4): geometrical means of volume size.

Columns (2) and (5): geometrical variability coefficients.

Columns (3) and (6): geometrical means of the structure sizes in percentage of the geometrical mean of the total brain volume (=relative value).

Column (7): percentage differences in structure size (sizes of the domestic pigs brain regions = 100%).

the hippocampus region of the wild boars is to a remarkably greater degree more variable in size than the hippocampus formation of the domestic pigs; the same holds true for a comprehension of all allocortical limbic structures, for the basolateral group and even for the total amygdaloid complex.

This conformity in the mentioned brain structures of wild boars and feral pigs in relation to the domestic pigs is remarkable, and will be further dealt with in the discussion, as also the possible importance of the partly considerable size differences of various structures despite equality of total brain sizes between feral and domestic pigs.

Discussion

The return of domestic pigs to a wild-life state is, like the process of domestication, a great experiment with animals, concerning which, unfortunately, a detailed report is lacking. Due to the special conditions of domestication, the brain weight has decreased by 33% from the wild boars to the domestic pigs. Hereby the sizes of partial structures of the central nervous system have been changed not homogeneously but very dissimilarly. These dissimilar changes in size correspond to heterogeneous changes in the performances of the sense organs and the behaviour of domestic animals (Herre and Röhrs, 1973). Under these circumstances the examination of the brains of feral domestic pigs has been of the utmost interest. The ascertained fact that these animals have the same brain sizes as domestic pigs is most remarkable. Recently, this finding could also be established for other domestic animals which have been feral throughout many generations; for instance, feral dogs, cats, goats, donkeys (Herre and Röhrs, 1973; Kruska, 1973a), and especially by the example of the dingo from Australia and New Guinea (Schultz, 1969), which has probably lived as a feral dog in free hunting-ground for the longest time. Accordingly the influence of the feral state on the brain size of the animals has so far not turned out to be reversible in relation to the influence of the domestication.

Nevertheless, certain differences could be ascertained in the quantitative structure of the brains of feral pigs in relation to the domestic pigs. Since there is no recorded report on the process of the return to feral state we have only little or no knowledge concerning the introduced starting-point-population of the Galapagos pigs. Apparently, marines introduced domestic pigs onto the Galapagos Islands approximately 70 to 140 years ago. They were supposed to breed in order to serve as a fresh meat reserve. With a certain reliability it can be assumed that the original population of the feral pigs belonged to a so-called primitive breed which at that time was more widely dispersed but which even today, at many places in the world, is kept under more or less extensive domestic conditions. However, so far, we have no idea whether significantly different brain compositions between different breeds of domestic pigs appear. Neither do we know whether, in the brain composition, extensive domestication conditions differ from the more intensive.

The basis of comparison is formed by brains of domestic pigs belonging to highly domesticated middle European breeds which we cannot accept as an original ancestral population of the Galapagos feral pigs. Therefore the shown deviations in the brain composition of feral pigs are not variations caused by the return to wild-life state but are, in fact, differences. We do not assume that fundamental variations, considering feral pigs in relation to domestic pigs, have taken place. The return to wild-life state is a process of selection of those domestic animals which still have the best chances to survive in free ground. This could probably be supported by the fact that animals from primitive breeds became feral as they adapted themselves more readily to wild-life conditions. In this connection it should be emphasised that the feral pigs are lean, muscular animals with relatively long legs and snouts and consequently resemble European boars in their bodily structure. However, in coloration they resemble domestic pigs.

The size differences in the structures between the brains of feral and domestic

pigs are insofar unimportant as they by no means correspond to the differences between European boars and domestic pigs. This has already been indicated by the identical total brain size. But some differences are still remarkable. As the selection principles in the free ground are different from those in domestication, an interpretation of the composition differences between brains of domestic pigs and those of the feral pigs according to functional aspects is to be attempted. It must be emphasized, however, that such an interpretation can only be an estimation, as the manifold performances of the various brain regions are known to only a limited degree. In particular, the five fundamental brain parts comprise a set of individual functions, which makes a detailed interpretation difficult.

According to our investigations, the feral pigs have a larger medulla oblongata than the domestic pigs, and this may be a proof of the doubtless stronger requirements for the movement of animals in free ground. But the finding on the cerebellum stands in contrast to this fact. This brain part, which is very important for the coordination and regulation of the movement and which is responsible for the spatial, postural adjustment of the animals, is surprisingly smaller in the feral pigs than in the domestic pigs. Thus an elucidation of this contradiction is not possible at the moment. But when we assume that among the feral pigs selectivity has favored those animals which had relatively large centers of the sense organs such as nose, eye and ear, then the larger medulla oblongata of these animals' descendants may possibly have been caused by the more strongly developed stato-acoustic centers in this brain part. The insignificantly larger diencephalon of the feral pigs can likewise be attributed to larger visual and acoustic centers.

Furthermore, the results achieved in the forebrain seem especially interesting. Although the telencephalon of the feral pigs is practically of the same size as that of the domestic pigs, a thorough analysis shows that the relative small part of the allocortex is 11% larger in the feral pigs. This result is interesting because two systems with different main functions within the allocortex can be contrasted (Hassler, 1964; Stephan, 1964): 1. the sense-dependent olfactory system and 2. the sense-independent limbic system. It is a well-known fact that, functionally as well as histologically, these systems are not homogeneous. The structures forming these systems each have different tasks. Also the manifold connections with other brain parts and the different structure through which they can be separated histologically could serve to explain the abovementioned fact. However, the basic function of the olfactory system is olfaction, and from the size of the primary olfactory center, the olfactory bulb, we can recognize a graduator in the performance of olfaction, because all olfactory impulses, starting from the nasal mucous membrane, are conveyed to the brain through this. But it still remains uncertain for which partial functions of olfaction the secondary olfactory centers of the palaeocortex are responsible. Feral pigs have a bulbus olfactorius 28% larger than that of the domestic pigs. Because pigs as macrosomatic animals are primarily orientated olfactorily, we can suppose that only those animals were selected which had sufficiently large olfactory centers to promote survival in natural environments. Compared with the olfactory bulbs of European wild boars, those of the feral pigs are approximately of the same size. This finding points to a good olfactory performance of the pigs which may be connected with food search, food examination, orientation, recognition of individuals and finding of sexual

partners under wild-life conditions. The observed behaviour pattern of the feral pigs on the Galapagos Islands points to these conclusions also.

In contrast to the primary olfactory center, the totality of the secondary centers is equally large in feral and domestic pigs. Also the various palaeocortical centers of the olfactory system must be regarded as being of the same size; only the tuberculum olfactorium is 19% smaller in the brains of the feral pigs; the basal nuclei are 25% larger. An interpretation of this finding is not possible at the moment.

We can also suppose the various structures of the limbic system in the allocortex to consist of manifold partial structures, although very little is known about the relationship between structure and function. After all, the hippocampus formation represents the center of this system and endogenous impulse production can with some certainty be regarded as being its basic and main function. Such endogenous performances as motivation, activity, aggressivity, attention as well as emotional reactions are of paramount importance to animals living in natural environments, for the maintenance of their position. When comparing the sizes of the brains of European boars and domestic pigs the largest domestic-dependent reductions were ascertained in respect of the hippocampus formation. We have considered this to be linked with the suppression of the aggressive instinct and with adjustment to life in the "entspannten Feld" (unstressed environments) of domestication (Kruska and Stephan, 1973; Röhrs and Kruska, 1969). Mainly for this reason, we have pointed out the possibility of recognizing, in the degree of reduction of the hippocampus formation, a graduator for the estimation of the domestication intensity (Kruska, 1973b). This conception finds evident verification in the results on brains of dogs (Schleifenbaum, 1973) and sheep (Ebinger, 1973).

Also, on the Galapagos Islands, the feral pigs do not live in an "entspanntem Feld". They have to hold their position in the natural environment to a greater extent than do domestic pigs, all the more, since, although natural enemies are lacking on these islands, intensive hunting is carried out by human beings. The evidently larger hippocampus formation of the feral pigs in relation to domestic pigs may be an indication of larger motivatory and activity functions and of stronger emotional reactions, too, in conjunction with aggression. Therefore it can be assumed that on the Galapagos Islands a selection of pigs with a larger hippocampus formation has resulted, even if this brain region still is about 30% smaller than in European boars.

In this connection, the lower variability in the size of almost all brain structures of the feral pigs, compared with the domestic pigs, is remarkable. This is also true for European boars, and once more we can ascertain that the individual variability of the brain composition has become more limited through natural selection than through domestication. However, the hippocampus formation seems to be an exception to this rule, because it shows larger individual variability in size both in European boars and feral pigs than in domestic pigs. The inferior variability of the limbic main center in domestic pigs can probably be ascribed to the artificial selection of animals subjected to almost no aggression by man.

Supplementary to our study, it would be interesting to examine brains of other feral pigs as, for instance, those in Australia, which have been living under

wild-life conditions for almost the same period of time (Pullar, 1950) as the Galapagos pigs.

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